

Jiaqi Zhu, Henry He

Professor Guodong Chen

Corporate Finance session 001

18 March, 2025

American Chemical Case Report

1. Which firms are the "identical twins" of the Collinsville investment? Using the 's for those assets and the methodology learned in this course, determine the appropriate discount rate for the Collinsville investment.

To identify the "identical twins" of the Collinsville Plant, we need to analyze its key characteristics and compare them with other companies in the industry. The Collinsville Plant specializes in sodium chlorate production, primarily serving the pulp and paper industry. It has a 40,000-ton production capacity, operates using graphite electrodes, and faces significant power costs, which account for 55-60% of manufacturing expenses. The plant is located in the Southeastern U.S., where the majority of sodium chlorate demand is concentrated. Also, from Exhibit 6 we can get the financial statements of the Collinsville Plant.

To find comparable companies, we can focus on several factors:

1. **Industry & Product Match** – Companies that produce sodium chlorate or supply chemicals to the pulp and paper industry.
2. **Production Capacity & Scale** – Firms with a similar plant size (e.g. around 40,000 tons annually).
3. **Cost Structure** – Companies with comparable power dependency and manufacturing cost breakdown.
4. **Geographic Location** – Firms operating in the Southeastern U.S. due to freight cost considerations.

From Exhibit 4, we can identify the domestic producers of sodium chlorate, along with their respective capacities and locations. These companies all satisfy the first factor—being part of the sodium chlorate industry—allowing us to narrow our focus to the remaining criteria: production capacity, cost structure, and geographic relevance.

After a comprehensive evaluation, we find that:

- Hooker Chemical Corporation
- Pennwalt Corporation

- Kerr-McGee Corporation

are the most relevant comparisons. Hooker Chemical's Columbus, MS (65,000 tons) and Taft, LA (40,000 tons) plants, Pennwalt's Calvert City, KY (37,000 tons) plant, and Kerr-McGee's Hamilton, MS (33,000 tons) plant all operate at a similar scale to Collinsville and serve the Southeastern U.S. market, ensuring similar freight cost considerations. Additionally, these companies are part of large diversified chemical corporations, meaning they likely share similar cost structures, especially in terms of power dependency and capital expenditures.

However, we found that there's no financial statements about Hooker Chemical Corporation in any of the tables. Thus we conducted additional searching. The company went out of business in 1968 due to an unusually high incidence of leukemia, birth defects, and other injuries reported by residents near its chemical waste disposal site (from Wikipedia). Therefore, we excluded Hooker Chemical Corporation.

Therefore, we conclude that **Pennwalt** and **Kerr-McGee** are the best candidates as "identical twins" for the Collinsville Plant.

From Exhibit 5 we can get the financial information of these two companies. We select the equity beta values and the proportion of equity and debt in 1978, as we believe that the most recent year's equity and debt values are the most informative. Also, we assume that these companies have very little chance of default, therefore the appropriate debt beta is zero. We can calculate each asset beta for each company by using this formula:

$$\beta_A = \beta_E E/(D+E)$$

and then take the average of the two asset beta.

Firm	Beta	Debt(D)	Equity(E)	Asset Beta
Pennwalt	1.33	31	69	0.9177
Kerr-McGee	1.06	17	83	0.8798

Thus the beta is about 0.89875. From page 4, we can find the risk-free rate, which is 9.5% (Long-term Treasury bonds). For the risk premium, a common estimate for the U.S. market risk premium in the late 1970s, including 1978, is around 6% to 8%. Given the high inflation and market conditions of that period, many financial analysts assume a market risk premium of approximately 7% when performing calculations related to that era. Thus we just use 7% as the risk premium. By calculation using the formula of CAPM, the appropriate discount rate for the Collinsville investment is **15.79%**.

2. Calculate the net present value of the Collinsville plant without the laminate technology. Remember that the transaction takes place at the end of 1979. Assume:

- **Project lasts 10 years (1980-1990)**
- **Project is 100% equity financed**
- **At the end of 10 years, the plant will be scrapped and will have zero salvage value**
- **For capital expenditures (1980-84) you should remember that $PPE_t = PPE_{t-1} + CapExt - Depr_t$ where PPE is property, plant and equipment**
- **Capital expenditures=600 per year for 1985-1989**
(Remember capital expenditures are depreciated)
- **Straight line depreciation**
- **Net working capital=accounts receivable+inventory-accounts payable**
- **No change in net working capital, 1985-89**
- **No change in operating profits, 1985-89**
- **Corporate tax rate = 48%**
- **debt = 0**

First, we sort out the CAPX over the decade. For 1980 to 1984, we use the formula in the assumption to find:

$$CapExt_t = PPE_t - PPE_{t-1} + Depr_t$$

From Exhibit 8 we can find the data of PPE as well as Depr from 1980 to 1984. Through calculation, the CAPX from 1980 to 1984 is respectively 485, 525, 560, 600 and 600 ('000s.)

	Expected 12/31	For the Years Ended December 31,				
	1979	1980	1981	1982	1983	1984
Depreciation		\$1,060	\$1,110	\$1,160	\$1,210	\$1,270
Net PP&E	10,600	10,025	9,440	8,840	8,230	7,560
CAPX		485	525	560	600	600

Then, according to the assumption, capital expenditures are 600 per year for 1985-1989, we got the full CAPX values over ten years.

Since the transaction takes place at the end of 1979, combining the data at the bottom of Exhibit 8, we can conclude that the initial expense at the end of 1979 is \$10.6 million.

In terms of the depreciation expense, we will follow the straight line depreciation, and we can get the annual depreciation expense from 1980 to 1984 in Exhibit 8. Since from 1985 there is CAPX, the depreciation expense should increase, by about \$60 thousand per period. And the salvage value in 1989 is about \$2640 thousand by calculation.

In addition, according to the assumption:

networking capital = accounts receivable + inventory - accounts payable

, we extract the corresponding value from Exhibit 8 and calculate the NWC value from 1980 to 1984.

	Expected	For the Years Ended December 31,				
	12/31 1979	1980	1981	1982	1983	1984
Accounts receivable	\$1,622	\$1,328	\$1,680	\$1,976	\$2,136	\$2,303
Inventories	651	598	756	889	961	1,036
Accounts payable	873	730	924	1,087	1,175	1,267
NWC	1400	1196	1512	1778	1922	2072

And from the assumption that there's no change in net working capital from 1985 to 1989, we can make the Change in NWC to be 0.

Also in Exhibit 8, we extract the operating profits data from 1980 to 1984. Question 2 also mentions that there is no change in operating profits from 1985 to 1989, so we assume that the profits from 1985 to 1989 are the same as in 1984, both being \$3026 ('000s.)

Then, using the formula of EBIT, we can get the EBIT result for each year.

Since the corporate tax rate is 48%, we can multiply the tax rate by the EBIT and get Taxes. Finally, according to the formula:

$$\text{NCF} = \text{EBIT} - \text{Taxes} + \text{Depreciation} - \text{CAPX} + \text{Change in NWC}$$

, we get the NCFs for each year. Last but not least, we use the discount rate found in question one, which is 15.79% to calculate the net present value of the Collinsville plant without the laminate technology today, which is **-\$3796.821 ('000s.)**

Year	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989
Initial Expense	-10600										
Depreciation		-1060	-1110	-1160	-1210	-1270	-1330	-1390	-1450	-1510	-1570
Salvage Value											2640
Operating Profit		833	2218	3035	3134	3026	3026	3026	3026	3026	3026
CAPX		-485	-525	-560	-600	-600	-600	-600	-600	-600	-600
NWC	1400	1196	1512	1778	1922	2072					
Change in NWC	19	204	-316	-266	-144	-150	0	0	0	0	0
EBIT		-227	1108	1875	1924	1756	1696	1636	1576	1516	1456
Corporate tax rate		48%	48%	48%	48%	48%	48%	48%	48%	48%	48%
Taxes		-108.96	531.84	900.00	923.52	842.88	814.08	785.28	756.48	727.68	698.88
NCF	-10581	660.96	845.16	1309	1466.48	1433.12	1611.92	1640.72	1669.52	1698.32	4367.12

3. Calculate the incremental NPV from adding the laminate technology to the Collinsville plant. What is the NPV of the Collinsville plant with the laminate technology? Assume:

- Without laminate, graphite costs would increase by 5% per year after 1984
- Save 15-20% of power costs with laminate.
- Power costs rise at 12% after 1984

According to the Pro Forma Financial Statement made by Dixon(Exhibit 8), its unlaminated graphite electrodes have the potential to profit in the future. It is analyzed to be realizable when applying laminate technology, which is a tool to effectively eliminate graphite costs and is expected to reduce power needs, about 15 to 20% according to the assumption. This is useful for Dixon Corporation since graphite costs would increase by 5% and power costs were projected to increase at 12% per year after 1984, while Dixon's plants depend largely on power. Thus, it can be observed that the NPV will increase after applying the laminate technology.

In terms of the new operating profit, we can add the saving cost back to the old operating profit, and the profits keep increasing thereafter. Assume we cut the original power cost by 17.5% and save the entire graphite cost. Since the company started using this technology from December of 1980, we take the cost saving into account from 1981.

From page 3 paragraph 6 we can get the data of initial outlay and the new depreciation expense. The cost of installing this disposable laminate is \$2.25 million, which can be considered as the newest initial outlay. After finding the initial outlay, we again use the straight-line depreciation method to get the annual depreciation of \$225 ('000s.) since the installation of the laminate technology at the Collinsville was scheduled for the end of 1980 instead of the first year, so there's a salvage value of 225 in the last year, in 1989. Then we use the same method in question 2 to calculate the EBIT and NCF.

Year	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989
Initial Expense	-10600										
Depreciation		-1060	-1110	-1160	-1210	-1270	-1330	-1390	-1450	-1510	-1570
Salvage Value											2640
Operating Profit		833	2218	3035	3134	3026	3026	3026	3026	3026	3026
CAPX		-485	-525	-560	-600	-600	-600	-600	-600	-600	-600
NWC	1400	1196	1512	1778	1922	2072					
Change in NWC	19	204	-316	-266	-144	-150	0	0	0	0	0
EBIT		-227	1108	1875	1924	1756	1696	1636	1576	1516	1456
Corporate tax rate		48%	48%	48%	48%	48%	48%	48%	48%	48%	48%
Taxes		-108.96	531.84	900.00	923.52	842.88	814.08	785.28	756.48	727.68	698.88
NCF	-10581	660.96	845.16	1309	1466.48	1433.12	1611.92	1640.72	1669.52	1698.32	4367.12
Graphite Cost		645	791	875	940	992	1041.6	1093.68	1148.36	1205.78	1266.07
Power Cost		6304	7735	9386	10526	11780	13193.6	14776.83	16550.05	18536.06	20760.39
Save Graphite Cost			791	875	940	992	1042	1094	1148	1206	1266
Save Power (17.5%)			1353.625	1642.55	1842.05	2061.5	2308.88	2585.95	2896.26	3243.81	3633.07
New Operation Profit		833.00	4362.63	5552.55	5916.05	6079.50	6376.48	6705.63	7070.62	7475.59	7925.14
New Initial Expense		-2250									
New Depreciation			-225	-225	-225	-225	-225	-225	-225	-225	-225
New Salvage Value											225
EBIT		-227	3027.625	4167.55	4481.05	4584.5	4821.48	5090.63	5395.62	5740.59	6130.14
Corporate tax rate		48%	48%	48%	48%	48%	48%	48%	48%	48%	48%
Taxes		-108.96	1453.26	2000.42	2150.90	2200.56	2314.31	2443.50	2589.90	2755.48	2942.47
NCF	-10581	-1589.04	2068.37	2726.13	3021.15	3128.94	3462.17	3662.13	3880.72	4120.11	7247.67

Thus the NPV of the Collinsville plant with the laminate technology is about **\$40.251 ('000s.)**

4. Under these terms, should Dixon purchase the Collinsville plant? Do you believe the profit projections given in the case? How would this affect your decision to purchase the plant?

Dixon should consider seriously whether to purchase the Collinsville plant or not because with a closer look and research we are able to see that the sales and the overall positive NPV in the end might be misleading due to the fact that it lack multiple factors that affect the sales structure which is an important factor when assessing the overall risk and value of the investment.

Firstly, the Collinsville plant's main source of revenue relies solely on the sales from sodium chlorate, a chemical which is primarily used in paper and pulp industries within the southern part of the United states. For example, The Collinsville plant has a production capacity of 40,000 tons and obtain nearly all of its sodium chlorate sales from the Southeastern region of the US, according to Exhibit 3, it accounts for roughly two-thirds of the total 435,000 tons of US sodium chlorate sales in 1979. This data can supports the mean of its overly high concentration on one single product and industry which may lead to higher risk factors where if the paper and pulp industries faces downturn or any possible governmental or state regulatory policies affects the sodium chlorate demand it would have a devastating impact on the plant's sales and revenue gain. With this factor being taken into account, sometimes when "Twin firms" or similar companies report similar statistics of sales, its revenue may not be solely from a single product but instead many different ones. Which is smarter when it comes to dealing with a specific shock to a singular income source. An example of this can be demonstrated by Pennwalt's Calvert City plant whose capacity is 37,000 tons and its sales is only one of several sales channels, which suggests its revenue base is more diversified, therefore able to reduce its risk.

Furthermore, let's take into consideration the geographic influence that may also have an impact on the market. From what we know, the Collinsville plant's operations are more constricted to the southern part of the United States because of the freight cost (800 Miles away of their principle market) consideration and being close to the large paper and pulp manufacturing industries. As mentioned before, this single concentrated geographical location increases its vulnerability to the regional economic fluctuations and local regulatory changes, and being compared with other companies with similar sales, they might have their geographical location diversified therefore bypassing this local market risk. For example Pennwalt operates three plants with significantly different regional locations each being Calvert city (37,000 tons), Portland, OR (26,000 tons) and Tacoma, WA. (9,000 tons) This spread is across more of the midwest and the pacific which strictly implies the differences that Pennwalt impose despite having similar sales structures.

Moreover, considering in the fact that Collinsville plant is sensitive to power cost which makes up around 55% to 60% (7000 KWH) of its manufacturing cost means that the rise of its energy cost would significantly decrease its profit margins, factor in this fact, companies with similar sales would have different capital and cost structure such as different distribution of energy source and operation efficiencies which can have different profitability. Broader of provider and sales mix can help absorb this potential cost increase if it doesn't affect the overall revenue stream equally.

Beyond just the systematic risk factors, let's dive more into the relationship between consumer and the product line. With further research, within the S&P global rating it can be seen that more well developed and successful firms tend to have a stable cash flow due to its product mix and its diversified consumer base, therefore we can use this information to assess the Collinsville plant which almost entirely depends on the pulp and paper industry, this dependency on the singular industry as mentioned above, not only have its geographical disadvantages, but also considering it is within the same industry which the consumer base are also relatively similar, therefore it is more adverse when it comes to new competitions and the demand of pulp and paper.

In conclusion, the raw sales data of Collinsville Plant may be relatively similar and compatible to firms such as Pennwalt and Kerr-McGee, but the sales structure tells a different story. Collinsville's heavy reliance on a single product line, consumer base, and geographic location combined with cost structure that is very price sensitive due to its market fluctuations assures its disadvantages when compared to its twin firms. It makes it far more riskier than its similar counterparts with a more diversified revenue mix. This exposure to risk marks the problem of whether the Power plant is able to generate consistent stable long term cash flow and maintain its profitability even with its laminate technology in place.

Citation:

"Industrials: Key Credit Factors: Business And Financial Risks In The Branded Consumer Products Industry." S&P Global Ratings, S&P Global, 10 Sept. 2008, disclosure.spglobal.com/ratings/pt/regulatory/article/-/view/sourceId/4891619.

2-Dec-15 Historical Corporate Top Tax Rate and Bracket, taxpolicycenter.org/sites/default/files/legacy/taxfacts/content/pdf/corporate_historical_bracket.pdf. Accessed 17 Mar. 2025.

Optional Question (Using APV):

Given: 20% of the enterprise's value is debt

$APV = NPV(\text{Base}) + NPV(\text{Financing})$

$D = 20\%$ $E = 80\%$

D/E (Debt to equity ratio) $= 0.2/0.8 = 0.25$

Since we already have the NPV_{Base} , and according to the APV we only need to add $NPV_{Financing}$, the present value of the interest tax shield, to get the APV. According to the formula:

$$V_{tax\ shield} = \sum_{t=1}^n \frac{T \times r_d \times D}{(1 + r_d)^t}$$

, we need T which is the corporate tax rate, r_d which is the debt rate of return, and we have to find the D, which is the Debt capacity ($V_{Unlevered}$) multiplied by debt rate which is 0.2.

T: Not explicitly provided in the article. We did some research online, and according to the *Historical Corporate Top Tax Rate and Bracket: 1909-2014*:

Year	Top Tax Rate	Top Tax Bracket
1980	46%	100,000
1981	46%	100,000
1982	46%	100,000
1983	46%	100,000
1984	46%	1,405,000
1985	46%	1,405,000
1986	46%	1,405,000
1987	40%	1,405,000
1988	34%	335,000
1989	34%	335,000

, we can find out that: 1980-1986: The top corporate tax rate was 46%. 1987: The top corporate tax rate was reduced to 40%. 1988-1989: The top corporate tax rate was further reduced to 34% under the Tax Reform Act of 1986. Then we assume using the arithmetic mean of those data, which is $46\% \times 0.7 + 40\% \times 0.1 + 34\% \times 0.2 = 43\%$, as our average corporate tax rate.

Debt rate of return (r_d): According to the question assumption, since the firm has AAA-rated bonds trading at 50 bps (0.50%) over Treasury bonds, and from the article page 4, the Treasury bond rate is 9.5%, we get: $r_d = 9.5\% + 0.50\% = 10\%$.

By applying these numbers, we can find out that the APV of Q2 is **-\$3997.457 ('000s)** and the APV of Q3 is **\$42.378 ('000s.)**